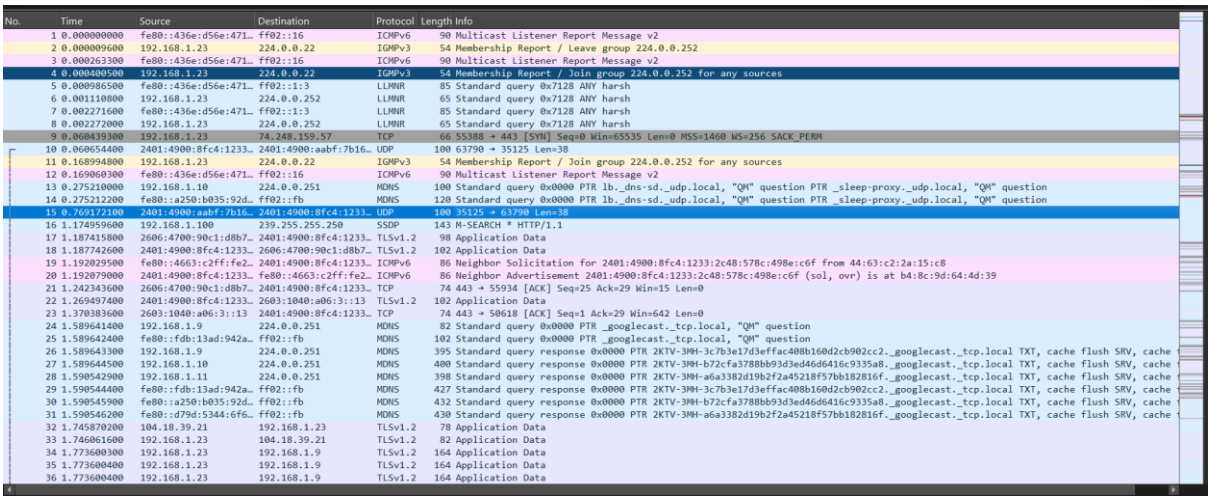
Software and Hardware Tools

1. Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

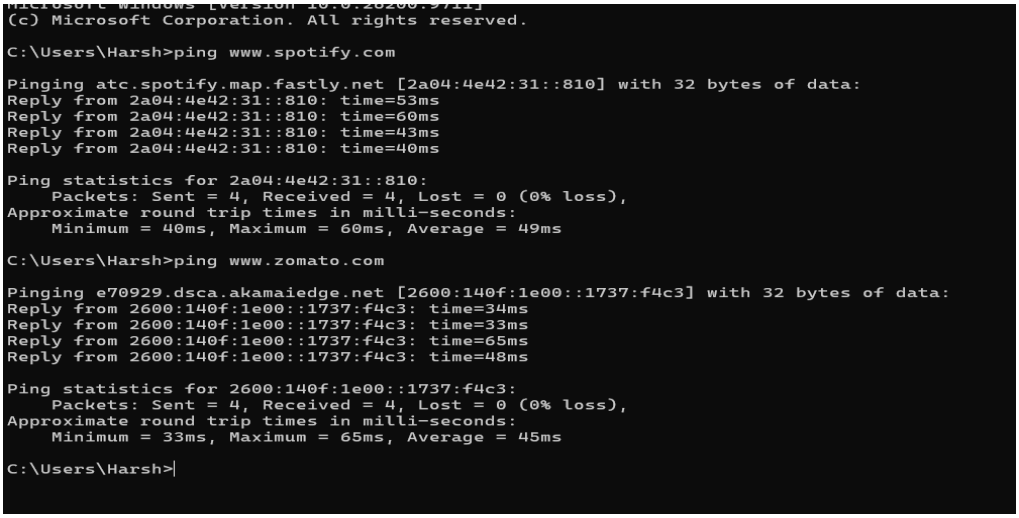


No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	fe80::436e:d56e:471…	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
2	0.000009600	192.168.1.23	224.0.0.22	IGMPv3	54	Membership Report / Leave group 224.0.0.252
3	0.000263300	fe80::436e:d56e:471…	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
4	0.000400500	192.168.1.23	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.252 for any sources
5	0.000986500	fe80::436e:d56e:471…	ff02::1:3	LLMNR	85	Standard query 0x7128 ANY harsh
6	0.001110800	192.168.1.23	224.0.0.252	LLMNR	65	Standard query 0x7128 ANY harsh
7	0.002271600	fe80::436e:d56e:471…	ff02::1:3	LLMNR	85	Standard query 0x7128 ANY harsh
8	0.002272000	192.168.1.23	224.0.0.252	LLMNR	65	Standard query 0x7128 ANY harsh
9	0.060439300	192.168.1.23	74.248.159.57	TCP	66	55388 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
10	0.060654400	2401:4900:8fc4:1233…	2401:4900:aabf:7b16…	UDP	100	63790 → 35125 Len=38
11	0.168994800	192.168.1.23	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.252 for any sources
12	0.169060300	fe80::436e:d56e:471…	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
13	0.275210000	192.168.1.10	224.0.0.251	MDNS	100	Standard query 0x0000 PTR lb._dns-sd._udp.local, "QM" question PTR _sleep-proxy._udp.local, "QM" question
14	0.275212200	fe80::a250:b035:92d…	ff02::fb	MDNS	120	Standard query 0x0000 PTR lb._dns-sd._udp.local, "QM" question PTR _sleep-proxy._udp.local, "QM" question
15	0.769172100	2401:4900:aabf:7b16…	2401:4900:8fc4:1233…	UDP	100	35125 → 63790 Len=38
16	1.174959600	192.168.1.100	239.255.255.250	SSDP	143	M-SEARCH * HTTP/1.1
17	1.187415800	2606:4700:90c1:d8b7…	2401:4900:8fc4:1233…	TLSv1.2	98	Application Data
18	1.187742600	2401:4900:8fc4:1233…	2606:4700:90c1:d8b7…	TLSv1.2	102	Application Data
19	1.192029500	fe80::4663:c2ff:fe2…	2401:4900:8fc4:1233…	ICMPv6	86	Neighbor Solicitation for 2401:4900:8fc4:1233:2c48:578c:498e:c6f from 44:63:c2:2a:15:c8
20	1.192079000	2401:4900:8fc4:1233…	fe80::4663:c2ff:fe2…	ICMPv6	86	Neighbor Advertisement 2401:4900:8fc4:1233:2c48:578c:498e:c6f (sol, ovr) is at b4:8c:9d:64:4d:39
21	1.242343600	2606:4700:90c1:d8b7…	2401:4900:8fc4:1233…	TCP	74	443 → 55934 [ACK] Seq=25 Ack=29 Win=15 Len=0
22	1.269497400	2401:4900:8fc4:1233…	2603:1040:a06:3::13	TLSv1.2	102	Application Data
23	1.370383600	2603:1040:a06:3::13	2401:4900:8fc4:1233…	TCP	74	443 → 50618 [ACK] Seq=1 Ack=29 Win=642 Len=0
24	1.589641400	192.168.1.9	224.0.0.251	MDNS	82	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
25	1.589642400	fe80::fdb:13ad:942a…	ff02::fb	MDNS	102	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
26	1.589643300	192.168.1.9	224.0.0.251	MDNS	395	Standard query response 0x0000 PTR 2KTV-3MH-3c7b3e17d3effac408b160d2cb902cc2._googlecast._tcp.local TXT, cache flush SRV, cache
27	1.589644500	192.168.1.10	224.0.0.251	MDNS	400	Standard query response 0x0000 PTR 2KTV-3MH-b72cfa3788bb93d3ed46d6416c9335a8._googlecast._tcp.local TXT, cache flush SRV, cache
28	1.590542900	192.168.1.11	224.0.0.251	MDNS	398	Standard query response 0x0000 PTR 2KTV-3MH-a6a3382d19b2f2a45218f57bb182816f._googlecast._tcp.local TXT, cache flush SRV, cache
29	1.590544400	fe80::fdb:13ad:942a…	ff02::fb	MDNS	427	Standard query response 0x0000 PTR 2KTV-3MH-3c7b3e17d3effac408b160d2cb902cc2._googlecast._tcp.local TXT, cache flush SRV, cache
30	1.590545900	fe80::a250:b035:92d…	ff02::fb	MDNS	432	Standard query response 0x0000 PTR 2KTV-3MH-b72cfa3788bb93d3ed46d6416c9335a8._googlecast._tcp.local TXT, cache flush SRV, cache
31	1.590546200	fe80::d79d:5344:6f6…	ff02::fb	MDNS	430	Standard query response 0x0000 PTR 2KTV-3MH-a6a3382d19b2f2a45218f57bb182816f._googlecast._tcp.local TXT, cache flush SRV, cache
32	1.745870200	104.18.39.21	192.168.1.23	TLSv1.2	78	Application Data
33	1.746061600	192.168.1.23	104.18.39.21	TLSv1.2	82	Application Data
34	1.773600300	192.168.1.23	192.168.1.9	TLSv1.2	164	Application Data
35	1.773600400	192.168.1.23	192.168.1.9	TLSv1.2	164	Application Data
36	1.773600400	192.168.1.23	192.168.1.9	TLSv1.2	164	Application Data

Top 3 Protocols to Look For

No.	Protocol	Purpose
1	DNS	Resolves domain names, such as instagram.com, to IP addresses.
2	TCP	Provides reliable data transmission for many internet connections.
3	TLS	Encrypts communication between your browser or app and secure servers.

2. Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.



```
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C:\Users\Harsh>ping www.spotify.com

Pinging atc.spotify.map.fastly.net [2a04:4e42:31::810] with 32 bytes of data:
Reply from 2a04:4e42:31::810: time=53ms
Reply from 2a04:4e42:31::810: time=60ms
Reply from 2a04:4e42:31::810: time=43ms
Reply from 2a04:4e42:31::810: time=40ms

Ping statistics for 2a04:4e42:31::810:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 40ms, Maximum = 60ms, Average = 49ms

C:\Users\Harsh>ping www.zomato.com

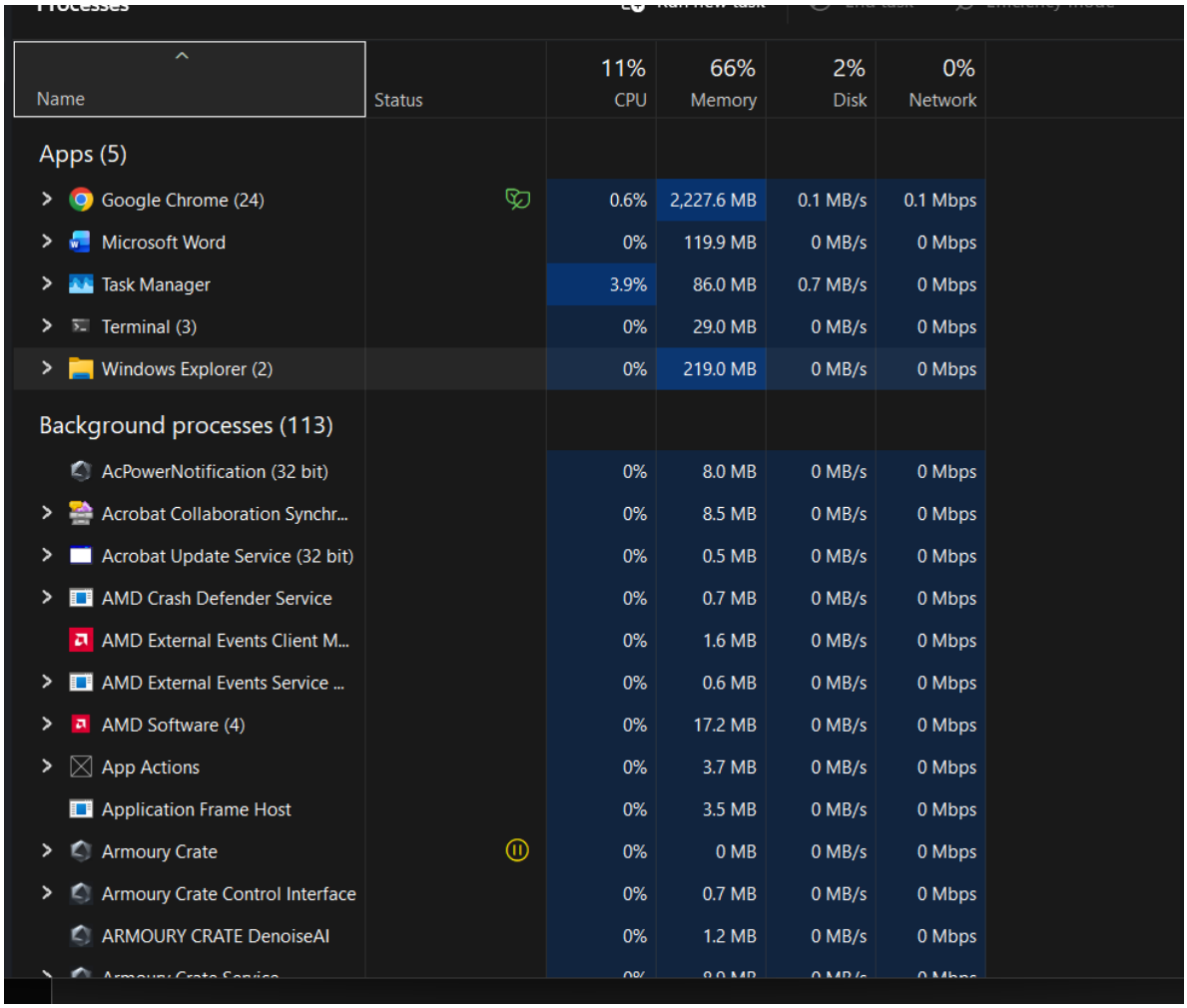
Pinging e70929.dsca.akamaiedge.net [2600:140f:1e00::1737:f4c3] with 32 bytes of data:
Reply from 2600:140f:1e00::1737:f4c3: time=34ms
Reply from 2600:140f:1e00::1737:f4c3: time=33ms
Reply from 2600:140f:1e00::1737:f4c3: time=65ms
Reply from 2600:140f:1e00::1737:f4c3: time=48ms

Ping statistics for 2600:140f:1e00::1737:f4c3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 33ms, Maximum = 65ms, Average = 45ms

C:\Users\Harsh>
```

Zomato responded faster on average, with a response time of 45 ms compared with Spotify's 49 ms. Both websites received all four packets successfully, with 0% packet loss.

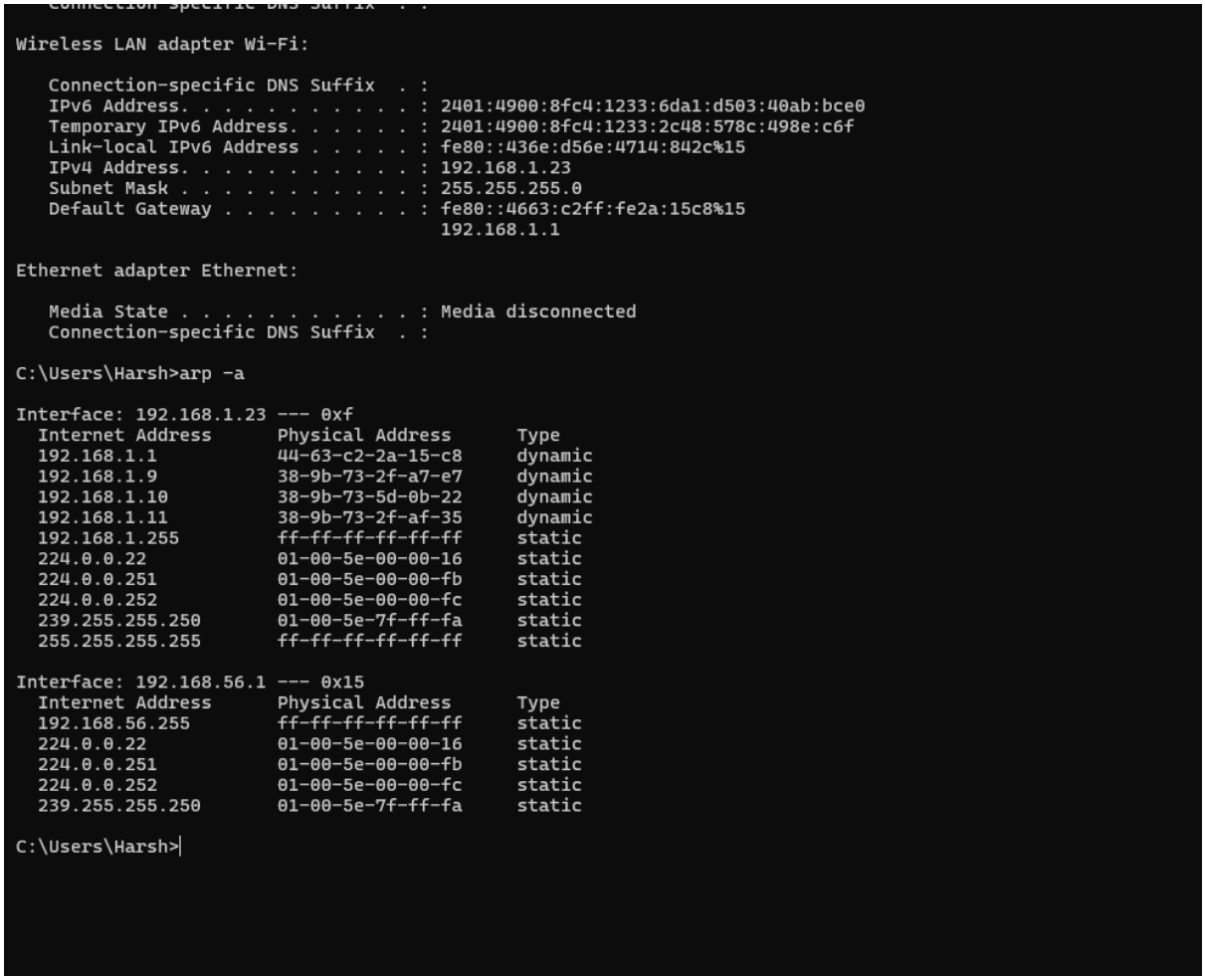
3. Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.



No.	Process / Application	Memory Usage
1	Google Chrome (24 processes)	2,227.6 MB
2	Microsoft Word	119.9 MB
3	Task Manager	86.0 MB
4	Terminal (3 processes)	29.0 MB
5	Windows Explorer (2 processes)	219.0 MB

Google Chrome is using the most memory among the visible applications, at approximately 2.23 GB. The screenshot does not show WhatsApp Desktop running, and the individual Chrome subprocesses are collapsed, so their separate memory usage cannot be identified.

4. Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network.



Result and Observation

The ipconfig and arp -a commands were executed in Windows Command Prompt to identify network configuration and discover devices on the local Wi-Fi network.

- **IPv4 Address:** 192.168.1.23
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.1.1

The arp -a command displayed the following discovered IP addresses and their MAC addresses:

No.	IP Address	MAC Address	Possible Device

1	192.168.1.1	44-63-C2-2A-15-C8	Wi-Fi Router
2	192.168.1.9	38-9B-73-2F-A7-E7	Connected device
3	192.168.1.10	38-9B-73-5D-0B-22	Connected device
4	192.168.1.11	38-9B-73-2F-AF-35	Connected device

Conclusion: The arp -a command identified the router and three additional IP-to-MAC address entries on the local network. The actual names of the other devices cannot be confirmed from this output alone.

5. Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

No.	Hardware Tool	When to Use It
1	Cable Tester	Used to check Ethernet cables for broken wires, incorrect wiring, and connectivity faults.
2	Crimping Tool	Used to attach RJ45 connectors to Ethernet cables when creating or repairing network cables.
3	Multimeter	Used to measure voltage, resistance, and continuity when troubleshooting electrical problems in networking equipment or power supplies.